DAURES GREEN HYDROGEN VILLAGE

University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use

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CO_2 Emission and Post-Combustion Carbon Capture Potentials for Otjikoto Biomass Power Station

BACKGROUND

Namibia's encroacher bush is an abundant biomass resource, but also a land-management challenge. On the other side, it also represents a large biogenic carbon stock with a potential to contribute to climate protection in various ways. The planned 40 MW Otjikoto Biomass Power Station (OBPS) is designed to utilize that biomass source for electricity generation and the encroacher bush will be harvested in around 100 km operating radius.

This factsheet presents some findings on the post-combustion CO_2 capture potential, that is a prerequisite for the layout and further use of the CO_2 in other processes e.g. for fuel production (i.e. SAF). The analysis considers the specific conditions of the Otjikoto's power plant site: a hot, arid environment with limited water availability. Key emission figures for the plant are summarised in Table 1. This pathway only represents one option to use biomass in a circular way for climate protection. For other options, e.g. via biochar please see factsheet 1.

CO_2 COMPOSITION OF VARIOUS GAS CURRENTS

The content of CO_2 and other gases determines the technical and economic effort to be made for separation of the flue gas. The flue gas from a biomass combustion power plant typically has a CO_2-content between 10 and 20%, N-content of 60-75 and around 2-8% of oxygen content (depending on air excess ratio).

The composition of CO2, N2, O2 N2 in various gas currents of biomass or biogas plants is given in Table 2.

Metric	Value
Fuel input (dry basis) [1]	26.89 t/h
Daily CO2 emissions	1120 – 1310 tCO2/day
Annual CO2 emissions	408,619 – 478,493 tCO2 /year
Flue gas CO2 concentration	13 – 15 vol%

Table 1. Key results for the Otjikoto Power Plant in a conservative maximum emissions scenario assuming full oxidation of fuel carbon

CARBON CAPTURE TECHNOLOGY

Various ways exist to capture carbon from a biomass power plant such as the OBPS (Fig. 1). CO_2 from the OBPS combustion plant must be separated and concentrated in a post-combustion carbon capture process from the flue gas stream.

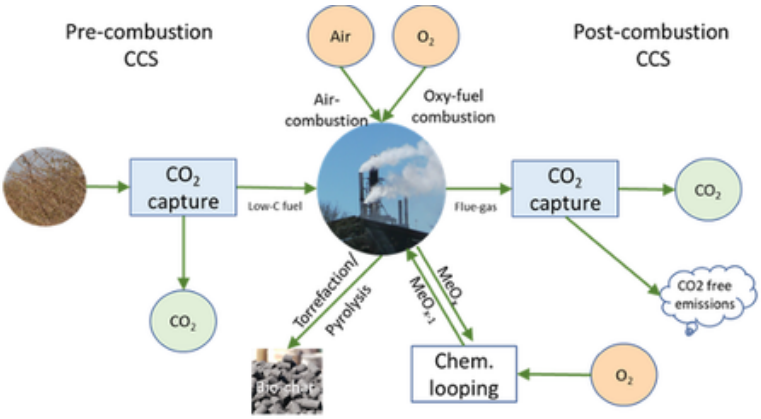

Figure 1: Principle pathways for Carbon Capture associated to a power plant like the OBPS. Pre-combustion pathways are suitable mainly for biogas power plants. Quality of CO2 in the flue gas is also related to the combustion process (e.g. air or O as oxidizing media)

Composition in %	CO2	O2	N2	H2	H2O	CH4
Dry air	0.042	20.9	78.1			
Standard flue gas	9.5		71.5		19	
Oxy-flue gas	60				40	
Biomass combustion	10-20	2-8	60-75		10-20	
Biogas	40					60
Hydrogen	20			80		

Table 2. Approximate average composition of different gas currents. Altered of Garcia et. al. [2]

CCS AND WATER CONSUMPTION

Due to the limited water availability in the hot arid climate evaporative cooling is not feasible and hybrid respiratory full dry cooling is the considered technology.

- For Hybrid configurations, water consumption can be as low as 3.4 m^3 of cooling water in Otjikoto Power Plant, assuming the process is well optimized and a flue gas condensate recovery rate of approximately 50%.
- ·Full air cooling minimises water use even more but increases auxiliary power demand (fans) and is sensitive to high ambient temperatures, potentially reducing overall plant efficiency.

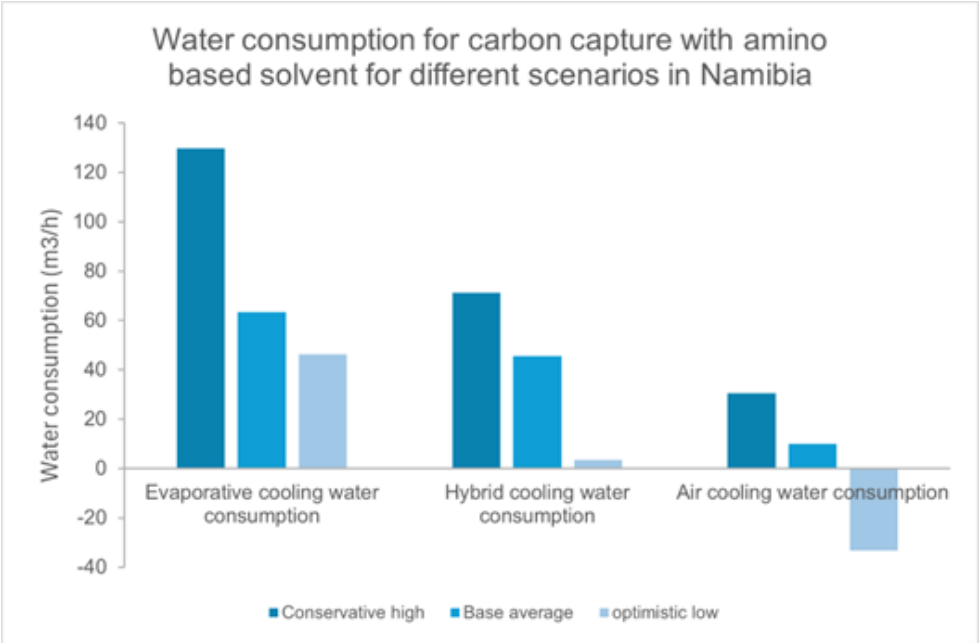

Figure 2: Water consumption for Amine-Based Carbon Capture under different cooling scenarios in Namibia project. Negative net water demand depends strongly on the water content in the biomass source and a high recovery of the condensate.

For CO_2-capture from flue gas of a power plant also various technologies exist. Generally absorption, adsorption, membrane based, cryogenic and chemical looping technologies are distinguished. The main post-combustion capture options relevant to the OBPS are compared in Table 3.

Technology	Suitability to Otjikoto Power Plant	Disadvantages	Water consumption	TRL
Amine absorption (hybrid/dry cooling)	· Most proven · Retrofit options · Handles 10 – 15% CO2 well · Lowest risk	· Energy penalty · dry cooling increases parasitic load · needs heat integration	Low to moderate	9
Calcium looping (CaL)	· Inherently dry · High-temperature operation suits hot climate · Abundant and cheap sorbent (limestone) · Biomass synergy	· Lower TRL · integration complexity; sorbent management	Low	6–7
Adsorption	· Modular and flexible design · Energy efficient · Lower regeneration energy	· Not yet fully commercial at large scale · Sensitive to flue gas conditioning (moisture)	Low	6–7
Membranes	· Electric, compact and simple · Near-zero process water footprint	· Multi-staging increases capex in energy for dilute streams (biomass) · Limited data on biomass combustion flue gas	Very low	6–8
Cryogenic (CCC)	· Solvent-free · High-purity and ready to transport CO2 · Some variants even recover water	· High refrigeration energy penalty exacerbated by hot ambient · High energy consumption for dilute gas	Low	6
Amine absorption (evaporative cooling)	· Conventional benchmark	· Water demand is incompatible with site constraints	High	9

Table 3. Carbon Capture technology assessment matrix [3], [4], [5], [6], [7]

RECOMMENDATION

For post-combustion CO_2 capture in the Namibian context, under water-constrained, hot arid conditions, **amine-based absorption with hybrid or dry cooling** is the top near-term option due to its TRL 9 maturity and proven low-water operation.

Technology selection is primarily driven by water demand and TRL, along with biomass flue gas synergies and CAPEX, placing calcium looping and adsorption as the second- and third-ranked low-water alternatives for future deployment.

ACKNOWLEDGMENTS

This fact sheet is designed for technical decision-makers (engineers, project developers, investors) while remaining accessible to policymakers. All figures are derived from conservative assumptions in the underlying technical assessment dated January 2026.

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PLEASE CONTACT: CRISTIAN ROJAS SALAMANCA

Institute of Energy Economics and Rational Energy Use, University of Stuttgart
cristian.rojas@st.ovgu.de

https://github.com/IER-Hy4Daures/Fact-Sheets

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